

TOPOLOGICAL SEMANTICS FOR SCOPED COMPUTATIONAL PATHS

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ABSTRACT. We give a topological semantics for a genuinely scoped computational-path rewrite presentation. A presentation consists of endpoint-changing geometric steps, coherent path realizations, and named endpoint-fixed rewrite rules. The rewrite relation is the least equivalence congruence generated by those rules and the explicit groupoid coherences; it is not ambient equality normalization.

For every presentation we construct a quotient arrow space and a canonical final-domain topological groupoid. Its multiplication is continuous on the quotient of the explicit composable-trace carrier, without any product-quotient assumption. We then prove an exact four-way classification of when this final domain agrees with the ordinary pullback topology. Thus the main groupoid theorem is unconditional, while the ordinary-pullback upgrade is a theorem about a separately checkable topological property, not an assumption hidden in the construction.

We also prove a positive ordinary-pullback theorem: compactness of the final composable domain and Hausdorffness of the ordinary domain force the two topologies to agree. A discrete finite-presentation corollary supplies an explicit class of presentations to which the ordinary upgrade applies.

The realization map to endpoint-fixed geometric homotopy classes is a continuous groupoid morphism. It is faithful exactly when the presentation is geometrically complete, and geometric completeness upgrades the comparison to a homeomorphism on arrow spaces. The universal presentation, whose rules are all endpoint-fixed homotopies, recovers the quotient-topologized fundamental groupoid. A concrete presentation on the unit additive circle has explicit zero, addition, and reversal rules, an inductive integer normal form, a continuous injection of the winding loop quotient, and a proved equivalence of that loop carrier with $\mathbb{Z}$. The actual torus $(\mathbb{R}/\mathbb{Z})^2$ is then classified by a two-coordinate winding theorem. The Lean development is an appendix and artifact: the mathematics below is independent of Lean.

1. INTRODUCTION

Computational paths were introduced to make equality witnesses explicit as traces of computational steps. The authors' program has developed this idea from propositional equality and identity types through path normalization, groupoid structure, and topological applications [dQdOR16a, RdQdO17b, RdQdOdV18, dVRdQdO25b, dVRdQdO25a]. This viewpoint is useful only while the trace syntax remains visible: if every proof of the same ambient equality is silently identified, then the intended

2020 *Mathematics Subject Classification.* Primary 03B38, 18N10, 54B30; Secondary 55Q05, 68V15.

Key words and phrases. computational paths, rewriting, fundamental groupoid, quotient topology, topological groupoid, formalized mathematics.

computational information has already been discarded. The central design problem is therefore to define a rewrite system whose semantic interpretation is sound without making semantic equality the rewrite relation itself.

The broader publication record includes the thesis and thesis abstract, the transport and univalence presentation, the 2021 calculations of fundamental groups and groupoids, and the Portuguese-language workshop paper on term conversion and homotopy [Ram18, Ram19b, RdQdO17a, dVRdQdO21, RdQdO21a, RdQdO21b]. The authors’ book gives the broader calculus and its historical context [RdQdOG26]. The earlier EBL and WoLLIC papers, the Springer monograph, and the fundamental-groups thesis are also part of that record [RdQdO14, RdQdO15, RdQdO16, RdQdO17c, dQdOR16b, Ram19a].

This paper develops the topological version of that design. We work with proof-irrelevant ambient paths, meaning ordinary continuous interval paths together with endpoint equalities, and with explicit finite geometric traces. The word “computational” refers to the trace and its declared rewrites; it does not refer to homotopy type theory, univalence, or a higher-inductive identity type. In particular, our fundamental-groupoid comparison is a semantic map out of a presented trace quotient, not an identification of proof-relevant identity types with topological spaces [HS98, RdQdO16, RdQdO17c, RdQdO15].

The contributions are the following.

- (i) We define continuous geometric rewrite presentations and a scoped rewrite relation generated by named rules, groupoid operations, and congruence. Soundness for geometric realization is proved by induction over the complete derivation system.
- (ii) We construct the quotient arrow space, its endpoint maps, identities, reversal, and a canonical final-domain topological groupoid. The final composable domain is the quotient of explicit composable traces, so the continuity proof does not use an unproved product of quotient maps.
- (iii) We prove an exact classification of the ordinary-pullback upgrade. Four conditions are equivalent: quotientness of the ordinary composable-pair map, continuity of the inverse final-to-ordinary identification, equality of the two composable-pair topologies, and quotientness of the raw ordinary-pair map. An open-map criterion is proved as a sufficient theorem.
- (iv) We prove a positive compact-Hausdorff compatibility theorem and its discrete finite-presentation corollary. Thus the paper contains an instantiated ordinary-pullback class, not only an exact obstruction.
- (v) We prove an effective normal-form criterion: a scoped normalizer that is geometrically separating yields geometric completeness. The criterion is instantiated by the finite-generator circle calculation.
- (vi) We prove functoriality for continuous presentation maps, a realization comparison and completeness criterion, and a universal presentation theorem. The comparison preserves the final composable-domain operation, not only a set-level map on arrows.
- (vii) We work out the actual additive circle from one oriented generator, with cancellation rules and a derived integer normal form. We then compute the genuine topological torus loop quotient as $\mathbb{Z} \times \mathbb{Z}$ by the product of the two circle covering classifications. Neither calculation makes a discrete-topology claim about the geometric loop spaces.

The paper is deliberately separate from the authors’ companion manuscript, “The Seifert–van Kampen Theorem via Computational Paths,” which is currently an unpublished arXiv preprint rather than a published paper [RdVdQdO25b]. That preprint studies a specific fundamental-group calculation. Here the principal result is the general topological construction for scoped rewrite presentations, together with the ordinary/final topology boundary and the universal semantic completion. The fundamental-groupoid terminology follows the standard topological treatment of groupoids and covering-space calculations [Bro06, Hat02].

To make the novelty boundary explicit, the earlier computational-path papers are used as foundations and comparisons, not silently counted as results of this manuscript. The 2016–2018 work develops the identity-type and explicit trace calculus; the 2021 papers calculate selected fundamental groups and groupoids; the accepted topological application and the weak-groupoid paper provide adjacent geometric and higher-structural results. The unpublished SVK manuscript is a companion preprint, not a published source and not a result claimed here. The present paper contributes the scoped topological quotient, the final/ordinary domain theorem, the positive compatibility class, the effective completeness criterion, and the circle–torus applications.

Record	Role in this paper	Result not imported from the record
Identity type and explicit paths [dQdOR16a, RdQdO17b, RdQdOdV18]	Syntax, equality background, and groupoid motivation	Scoped topological quotient and its two composable-domain topologies
2021 computational-path calculations [dVRdQdO21, RdQdO21a, RdQdO21b]	Earlier concrete algebraic and homotopical applications	Compact-Hausdorff compatibility and normal-form completeness
Topological and weak-groupoid papers [dVRdQdO25b, dVRdQdO25a]	Adjacent topological and higher-categorical context	Final-domain semantics and the exact ordinary/final comparison
SVK companion preprint [RdVdQdO25b]	Separate unpublished calculation	No theorem here depends on its proof or is presented as its result
This paper	Standalone topological semantics	Scoped quotient, positive topology class, effective completeness, circle, and genuine torus classification

TABLE 1. Theorem-level novelty and overlap boundary.

2. GEOMETRIC TRACES AND SCOPED PRESENTATIONS

Throughout, X is a topological space. Let E be a topological space of primitive steps, with continuous endpoint maps $s, t : E \rightarrow X$ and a continuous family of realized interval paths

$$\rho(e) : s(e) \rightsquigarrow t(e).$$

The continuity requirement means that the corresponding map into the compact-open path space X^I is continuous. The endpoint-indexed notation below is convenient for stating composition, but the underlying trace syntax is independent of a proof of ambient equality.

Definition 2.1 (geometric trace). The geometric trace inductive family is generated by

$$\begin{array}{c} \overline{\text{Trace}(a, a)}, \quad \frac{e : E}{\text{Trace}(s(e), t(e))}, \quad \frac{p : \text{Trace}(a, b) \quad q : \text{Trace}(b, c)}{p ; q : \text{Trace}(a, c)}, \\ \frac{p : \text{Trace}(a, b)}{p^{-1} : \text{Trace}(b, a)}. \end{array}$$

Its realization is defined recursively by constant paths, $\rho(e)$, concatenation, and reversal. The trace length $\ell(p) \in \mathbb{N}$ is defined by length 0 for reflexivity, length 1 for a primitive step, addition under concatenation, and preservation under reversal.

The topology on each trace fiber observes both $\ell(p)$ and the realized path. A coherent open path is a triple

$$u = (p, \gamma, H),$$

where $p : \text{Trace}(a, b)$, $\gamma : a \rightsquigarrow b$ is an arbitrary continuous interval path, and H is an endpoint-fixed homotopy from γ to the realization of p . Let T denote the endpoint-varying carrier of coherent open paths. Its topology observes source, target, trace length, realized trace, and chosen geometric representative. This choice is important: the topology remembers geometric motion while the quotient relation below controls which computational traces are identified.

Definition 2.2 (scoped presentation). A continuous geometric rewrite presentation $\mathcal{P}$ on (X, E) consists of a predicate

$$R_{\mathcal{P}}(p, q), \quad p, q : \text{Trace}(a, b),$$

and, for every instance of $R_{\mathcal{P}}(p, q)$, an endpoint-fixed homotopy between the realizations of p and q .

The semantic witness is part of the presentation data. No completeness condition is included in the definition.

Definition 2.3 (scoped rewrite equality). For parallel traces, write $p \simeq_{\mathcal{P}} q$ for the inductively generated relation whose constructors are reflexivity, the named generators $R_{\mathcal{P}}$, symmetry, transitivity, congruence under concatenation and reversal, and the following strict groupoid coherences:

$$\begin{array}{lll} (1_a); p \simeq_{\mathcal{P}} p, & p; (1_b) \simeq_{\mathcal{P}} p, & (p; q); r \simeq_{\mathcal{P}} p; (q; r), \\ p^{-1}; p \simeq_{\mathcal{P}} 1_b, & p; p^{-1} \simeq_{\mathcal{P}} 1_a, & (p^{-1})^{-1} \simeq_{\mathcal{P}} p, \\ (1_a)^{-1} \simeq_{\mathcal{P}} 1_a, & (p; q)^{-1} \simeq_{\mathcal{P}} q^{-1}; p^{-1}. \end{array}$$

Only the rules declared by $\mathcal{P}$ and these structural constructors are available. In particular, ambient equality of traces is not a constructor.

Theorem 2.4 (realization soundness). *If $p \simeq_{\mathcal{P}} q$, then the realized paths are endpoint-fixed homotopic.*

Proof. Induct on the derivation of $p \simeq_{\mathcal{P}} q$. Reflexivity is the constant homotopy, and a named generator is sound by the corresponding presentation witness. Symmetry

and transitivity use symmetry and vertical composition of endpoint-fixed homotopies. For concatenation, the two inductive homotopies are combined by horizontal concatenation; for reversal, reverse the homotopy parameter. The unit, associativity, cancellation, double-reversal, and reversal-of-composite cases are the standard endpoint-fixed homotopies of interval paths. Each case is therefore discharged by the corresponding constructor of the ambient homotopy equivalence relation. $\square$

Remark 2.5. The soundness theorem is one-way. Equality of realized geometric homotopy classes does not imply scoped rewrite equality unless a separate geometric completeness theorem is proved. This distinction is the logical reason that the presentation can retain computational information.

3. THE QUOTIENT ARROW SPACE

The scoped relation on endpoint-varying coherent paths compares endpoints and then compares their traces using the declared presentation. Write $u \approx v$ for this relation on T . The quotient arrow space is

$$G_{\mathcal{P}} = T / \approx,$$

with the quotient topology induced by $q : T \rightarrow G_{\mathcal{P}}$. Endpoint maps descend to continuous maps

$$s, t : G_{\mathcal{P}} \longrightarrow X.$$

Reflexive paths and reversal descend because the scoped relation is closed under the corresponding constructors.

The explicit composable carrier is

$$T^{(2)} = \{(u, v) \in T \times T : t(u) = s(v)\}.$$

Its quotient relation is componentwise scoped equivalence, and its quotient is denoted $G_{\mathcal{P}}^{(2), \text{fin}}$. Concatenation of representatives gives a well-defined map

$$m_{\text{fin}} : G_{\mathcal{P}}^{(2), \text{fin}} \longrightarrow G_{\mathcal{P}}.$$

The quotient topology is chosen precisely so that this map has the required continuity.

Theorem 3.1 (canonical final-domain groupoid). *For every continuous geometric rewrite presentation $\mathcal{P}$:*

- (a) s, t , the identity map $X \rightarrow G_{\mathcal{P}}$, and reversal $G_{\mathcal{P}} \rightarrow G_{\mathcal{P}}$ are continuous;
- (b) m_{fin} is continuous;
- (c) the descended operations satisfy the left and right unit laws, inverse laws, and associativity.

Thus $G_{\mathcal{P}}$ is a final-domain topological groupoid. The phrase “final-domain topological groupoid” records the domain on which multiplication is continuous.

Proof. Continuity of the endpoint maps and identities follows from the quotient criterion, because the corresponding raw maps are continuous. The same argument applies to reversal. The raw concatenation map $T^{(2)} \rightarrow T$ is continuous and respects the componentwise quotient relation, so it descends to m_{fin} ; continuity follows from the defining quotient topology on $G_{\mathcal{P}}^{(2), \text{fin}}$.

For the algebraic laws, choose representatives. Left and right units reduce to the scoped constructors $(1_a); p \simeq_{\mathcal{P}} p$ and $p; (1_b) \simeq_{\mathcal{P}} p$. The two inverse laws reduce to $p^{-1}; p \simeq_{\mathcal{P}} 1_b$ and $p; p^{-1} \simeq_{\mathcal{P}} 1_a$. Associativity reduces to $(p; q); r \simeq_{\mathcal{P}} p; (q; r)$. The quotient soundness criterion then turns each scoped derivation into equality

of quotient arrows. The result is independent of representative choice because concatenation was defined through the componentwise quotient. $\square$

4. ORDINARY AND FINAL COMPOSABLE-PAIR TOPOLOGIES

There is another natural composable domain. Let

$$G_{\mathcal{P}}^{(2),\text{ord}} = \{(x, y) \in G_{\mathcal{P}} \times G_{\mathcal{P}} : t(x) = s(y)\}$$

with the subspace topology. The underlying set of this space is the same as the underlying set of the final domain, but its topology need not be the same. The canonical map

$$q^{(2)} : T^{(2)} \longrightarrow G_{\mathcal{P}}^{(2),\text{ord}}$$

is continuous and surjective. Product quotient maps are not quotient maps in arbitrary topological spaces, so quotientness of $q^{(2)}$ is an additional property, not a formal consequence of quotientness of q .

Definition 4.1 (product-quotient compatibility). The presentation has product-quotient compatibility if $q^{(2)}$ is a quotient map onto the ordinary composable-pair space.

The final domain can be identified with the ordinary domain by the bijection that is the identity on the underlying set. One direction is always continuous; the reverse direction is the exact point at which compatibility is needed.

Theorem 4.2 (ordinary/final comparison). *For every presentation $\mathcal{P}$, the following are equivalent:*

- (1) *product-quotient compatibility holds;*
- (2) *the identity map from the ordinary composable domain to the final composable domain is continuous;*
- (3) *the ordinary and final composable-pair topologies coincide;*
- (4) *the raw ordinary-pair map, viewed from $T^{(2)}$, is a quotient map.*

When these conditions hold, multiplication is continuous from the ordinary pullback domain. No condition in this theorem is used by the canonical final-domain groupoid theorem.

Proof. The final topology is the coinduced topology for $q^{(2)}$; hence quotientness of $q^{(2)}$ is equivalent to equality between the final topology and the ordinary topology. Equality of the two topologies is equivalent to continuity of the identity in the reverse direction because the forward identity is always continuous. The raw ordinary-pair map is the composite of the raw composable quotient with the identity between the two pair domains. The composition and cancellation criteria for quotient maps therefore show that its quotientness is equivalent to quotientness of $q^{(2)}$. Finally, when $q^{(2)}$ is quotient, multiplication is continuous by the quotient criterion applied to its continuous representative-level concatenation. $\square$

Proposition 4.3 (open-map criterion). *If the bijection from the final composable quotient to the ordinary pair space is an open map, then product-quotient compatibility holds. In particular, an open quotient presentation of the composable domain supplies the ordinary pullback upgrade.*

Theorem 4.4 (compact-Hausdorff compatibility). *Suppose that the final composable domain $G_{\mathcal{P}}^{(2),\text{fin}}$ is compact and that the ordinary composable domain $G_{\mathcal{P}}^{(2),\text{ord}}$ is Hausdorff. Then product-quotient compatibility holds. Hence the final and ordinary composable-pair topologies agree and ordinary-pullback multiplication is continuous.*

Proof. The identity map

$$j : G_{\mathcal{P}}^{(2),\text{fin}} \longrightarrow G_{\mathcal{P}}^{(2),\text{ord}}$$

is a continuous bijection by the comparison theorem. A continuous bijection from a compact space to a Hausdorff space is a homeomorphism: its image of every closed set is compact, hence closed, so its inverse is continuous. The quotient map defining the final domain followed by j is therefore the raw ordinary-pair map followed by a homeomorphism, and is a quotient map. This is exactly product-quotient compatibility. $\square$

Corollary 4.5 (finite discrete presentations). *Let $\mathcal{P}$ be a scoped presentation whose arrow space is finite and discrete and whose explicit composable-trace carrier is discrete. Then the ordinary and final composable-pair topologies coincide.*

Proof. The quotient of a discrete space is discrete for the quotient topology, so the final composable domain is discrete. It is finite because its underlying set is a subset of the square of the finite arrow space. The ordinary composable domain is a finite subspace of a discrete square and is therefore Hausdorff. The compact-Hausdorff theorem applies. $\square$

The compact-Hausdorff theorem is the promised positive class: it establishes the ordinary-pullback upgrade from independently checkable properties of the actual presentation, rather than leaving the four-way criterion as the only result. The final-domain theorem remains the general statement and does not need these hypotheses.

5. FUNCTORIALITY

Let $\mathcal{P}$ and $\mathcal{Q}$ be presentations on (X, E) and (Y, F) . A presentation map consists of a continuous map $f : X \rightarrow Y$, a continuous step map preserving endpoints and realizations, and a proof that every named rule of $\mathcal{P}$ maps to a scoped derivation in $\mathcal{Q}$.

Theorem 5.1 (functoriality). *Every presentation map $F : \mathcal{P} \rightarrow \mathcal{Q}$ induces a continuous map*

$$F_1 : G_{\mathcal{P}} \longrightarrow G_{\mathcal{Q}}$$

that preserves source, target, identities, reversal, and final-domain composition. The assignment preserves identity maps and composites.

Proof. Induction on scoped rewrite derivations proves that the step map sends related traces to related traces. It therefore descends through the quotient. The raw map on coherent paths is continuous, so the quotient criterion gives continuity on arrows. Applying the step map to reflexivity, reversal, and concatenation gives the corresponding preservation equations before quotienting. The same argument on the explicit composable carrier gives a continuous map on final composable domains and proves final composition preservation. Identity and composite step maps are definitionally the identity and composite maps on raw traces; quotient extensionality completes the proof. $\square$

The same construction gives ordinary-pair continuity for every presentation map. If both source and target presentations satisfy product-quotient compatibility, the ordinary multiplication maps are continuous as well. This is a derived theorem, not the definition of functoriality.

6. GEOMETRIC COMPARISON AND COMPLETENESS

Let H_{geo} be the quotient of coherent open paths by equality of endpoints and endpoint-fixed homotopy of chosen geometric representatives. The realization comparison

$$R_{\mathcal{P}} : G_{\mathcal{P}} \longrightarrow H_{\text{geo}}$$

maps a scoped class to its geometric homotopy class.

Theorem 6.1 (comparison morphism). *The map $R_{\mathcal{P}}$ is continuous, surjective on the geometric quotient, and preserves source, target, identity, reversal, and final-domain composition. In particular it is a continuous groupoid morphism for the final-domain structures.*

Proof. Realization soundness makes the map well-defined. Its composite with the raw quotient map is the continuous geometric quotient map, so the quotient criterion proves continuity. Every geometric class has a coherent raw representative by definition, proving surjectivity. The operation equations follow from the recursive realization of reflexivity, concatenation, and reversal. For the final domain, map a strong pair to the corresponding strong pair of geometric classes; the quotient representative of its composition is the geometric composition of the two component representatives. $\square$

Definition 6.2 (geometric completeness). The presentation is geometrically complete if, for all raw coherent paths u, v , equality of their geometric quotient codes implies scoped equivalence:

$$R_{\mathcal{P}}(q(u)) = R_{\mathcal{P}}(q(v)) \implies u \approx v.$$

Theorem 6.3 (faithfulness and homeomorphism). *The comparison $R_{\mathcal{P}}$ is faithful if and only if $\mathcal{P}$ is geometrically complete. If $\mathcal{P}$ is geometrically complete, then*

$$R_{\mathcal{P}} : G_{\mathcal{P}} \xrightarrow{\cong} H_{\text{geo}}$$

is a homeomorphism on arrow spaces and an isomorphism of abstract groupoids.

Proof. If the comparison is injective, equality of geometric codes for two raw paths gives equality of their scoped quotient classes, which is precisely scoped completeness. Conversely, completeness turns equality of comparison images into equality of scoped classes. The comparison is always surjective and continuous. Under completeness, define the inverse by lifting a geometric class to a raw representative and taking its scoped class. Completeness makes this lift well-defined, and the quotient criterion makes it continuous. The two composites are identity maps by quotient induction. $\square$

Definition 6.4 (effective normal-form certificate). An effective normal-form certificate for $\mathcal{P}$ is a map

$$N : T \longrightarrow T$$

such that, for all raw coherent paths u, v ,

- (a) $u \approx N(u)$;
- (b) equality of geometric quotient codes implies $N(u) = N(v)$; and
- (c) $N(u) = N(v)$ implies $N(u) \approx N(v)$.

The first clause is a derivable normalization certificate, the second is semantic separation of normal forms, and the third records the scoped proof that equal normal forms represent the same presented arrow.

Theorem 6.5 (normal-form completeness criterion). *An effective normal-form certificate implies geometric completeness. Therefore the realization comparison is faithful and is a homeomorphism on arrow spaces.*

Proof. Assume that the geometric codes of u and v are equal. Clause (b) gives $N(u) = N(v)$, and clause (c) gives $N(u) \approx N(v)$. Combining this with clause (a) for u and the symmetric form of clause (a) for v yields

$$u \approx N(u) \approx N(v) \approx v.$$

This is precisely geometric completeness. The faithfulness and homeomorphism claims follow from the comparison theorem. $\square$

7. THE UNIVERSAL GEOMETRIC PRESENTATION

For every topological space X , take the primitive step space to be the continuous interval path space X^I . Its endpoint maps are evaluation at 0 and 1. Declare two parallel traces to be named generators exactly when their realized paths are endpoint-fixed homotopic.

Theorem 7.1 (universal completion). *For the universal presentation $\mathcal{U}_X$, scoped rewrite equality of parallel traces is equivalent to endpoint-fixed homotopy of their realizations. Consequently,*

$$Gu_x \cong \Pi_1^q(X)$$

as abstract groupoids, and the comparison is a homeomorphism on arrow spaces.

Proof. Soundness is the general soundness theorem. Conversely, an endpoint-fixed homotopy is itself one of the named universal generators, so it is a scoped rewrite in one step. The equivalence of quotient relations follows after including endpoint equality. The geometric completeness theorem then gives the homeomorphism and groupoid isomorphism. $\square$

Remark 7.2. The multiplication in this completion is understood on the final composable domain. This is the unconditional topological statement. For compact final composable domains with Hausdorff ordinary domains, the compact-Hausdorff theorem above identifies this multiplication with the ordinary pullback operation.

8. THE ADDITIVE CIRCLE: A FINITE GENERATOR AND ITS COMPLETION

Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ be the unit additive circle. We use the actual topological circle, not an inductive one-point model. The minimal presentation has one oriented primitive step a at the base point 0, with realization

$$a(t) = [t] \in \mathbb{T}.$$

The formal reversal supplies a^{-1} ; no integer-indexed primitive step is assumed. A based trace is therefore a finite signed word in a and a^{-1} . The only non-structural generators are the two cancellation rules

$$a; a^{-1} \simeq_{\mathcal{P}} 1_0, \quad a^{-1}; a \simeq_{\mathcal{P}} 1_0,$$

closed under the scoped congruence.

For a signed word w , let $\exp(w)$ be the number of positive letters minus the number of negative letters. Let c_n be the canonical word consisting of n copies of a when $n \geq 0$ and $-n$ copies of a^{-1} when $n < 0$.

Lemma 8.1 (finite-generator reduction). *Every signed word w has a finite scoped derivation*

$$w \simeq_{\mathcal{P}} c_{\exp(w)}.$$

Moreover, two canonical words are scoped equivalent only when their exponents are equal after geometric realization.

Proof. Scan the word from left to right while retaining a reduced signed word. If the next sign agrees with the last retained sign, append it. If the signs differ, delete the adjacent pair; this is one of the two cancellation generators, possibly whiskered by the already scanned prefix and the unscanned suffix. Each deletion strictly decreases the word length, so the procedure terminates. A terminal reduced word has no adjacent sign change and hence consists entirely of positive letters or entirely of negative letters. The integer $\exp$ is unchanged by every deletion, so the terminal word is exactly $c_{\exp(w)}$. The same invariant is the endpoint of the lift to $\mathbb{R}$; therefore endpoint-fixed homotopy of canonical words forces equality of their exponents. $\square$

The reduction lemma is the effective normal-form certificate from Section 4: the reduction derivation gives clause (a), the lifted endpoint gives clause (b), and equality of canonical words gives clause (c). Thus the minimal circle presentation is geometrically complete on its based loop fiber.

For comparison with the checked artifact, the completed integer presentation has a primitive step λ_n for each $n \in \mathbb{Z}$,

$$\lambda_n(t) = [tn].$$

Its named rules are the derived integer rules

$$(n; m) \simeq_{\mathcal{P}} (n + m), \quad 1_0 \simeq_{\mathcal{P}} 0, \quad (n)^{-1} \simeq_{\mathcal{P}} (-n).$$

The finite-generator argument above explains why this completion does not build the integer classification into the primitive syntax.

Let L be the quotient of coherent based circle loops by endpoint-fixed homotopy. The map sending a loop class to its winding number and the map sending n to the standard loop class are inverse maps.

Theorem 8.2 (circle winding classification). *The finite-generator scoped presentation is geometrically complete on the based loop fiber, and there is an equivalence*

$$L \simeq \mathbb{Z}.$$

The induced continuous map from L into the scoped arrow carrier is injective, and its image is exactly the source-target fiber over the base point. In particular the scoped presentation is nondegenerate: the images of the zero and one winding loops are distinct.

Proof. The quotient map to winding is well-defined because endpoint-fixed homotopies lift to homotopies with equal lifted endpoints. The standard loop of winding n has lift $t \mapsto tn$. Every based loop is homotopic to the standard loop with the endpoint of its lift, so the two maps are inverse. The scoped map is continuous by the quotient criterion. If two of its values agree, soundness gives a geometric homotopy between the two chosen loops; winding invariance then gives equality of their integer normal forms. The image condition is obtained by quotient induction on an arrow and by inserting the based loop representative when the endpoints are both 0. The zero and one classes are distinct because winding sends them to distinct integers. $\square$

The Lean declarations `circleBasedNormalForm`, `circleBasedNormalForm_scoped`, and `circleBasedNormalFormCertificate` record the same based completeness argument for the completed integer presentation. The equivalence $L \simeq \mathbb{Z}$ is a quotient classification, not a claim that L or the corresponding scoped fiber is discrete.

9. THE GENUINE TOPOLOGICAL TORUS

Let

$$\mathbb{T}^2 = (\mathbb{R}/\mathbb{Z}) \times (\mathbb{R}/\mathbb{Z})$$

with base point $(0, 0)$. Use two finite oriented generators

$$a(t) = ([t], [0]), \quad b(t) = ([0], [t]).$$

The scoped rules are the inverse cancellations for a and b , together with the commuting square

$$a; b \simeq_P b; a.$$

The commuting square is sound: it is the projection modulo $\mathbb{Z}^2$ of the evident square homotopy in $\mathbb{R}^2$. Closure under symmetry, reversal, and whiskering supplies the corresponding commuting rules for all four sign combinations.

For a signed word w , let $m(w)$ and $n(w)$ be its signed counts of a and b . Move every b -letter to the right using the commuting square and then perform the two cancellation algorithms from the circle. This is a finite terminating derivation

$$w \simeq_P a^{m(w)}; b^{n(w)}.$$

The pair $(m(w), n(w))$ is invariant under every scoped rule.

Theorem 9.1 (torus winding classification). *For the actual topological torus, the quotient $L_{\mathbb{T}^2}$ of based coherent loops by endpoint-fixed homotopy satisfies*

$$L_{\mathbb{T}^2} \cong \mathbb{Z} \times \mathbb{Z}.$$

The finite-generator scoped presentation is geometrically complete on the based loop fiber, and its realization comparison is injective there.

Proof. The universal cover is the product map

$$\mathbb{R}^2 \longrightarrow \mathbb{T}^2, \quad (x, y) \longmapsto ([x], [y]).$$

The lift of a based loop has endpoint $(m, n) \in \mathbb{Z}^2$, and endpoint-fixed homotopy preserves this pair. Conversely, the two coordinate projections of any based torus loop are circle loops; the circle lifting theorem gives endpoint-fixed homotopies to the standard loops of windings m and n . Taking their product gives a homotopy to $a^m; b^n$. Thus the coordinate winding pair classifies geometric loops. The word

reduction above gives the corresponding scoped derivation, so the normal-form criterion proves geometric completeness and injectivity. $\square$

This example is genuinely topological: the two coordinates are realized by the additive-circle covering map, and the product homotopy is proved at the level of continuous interval paths. It is not the older synthetic torus carrier and does not use the companion SVK preprint. The Lean artifact records the product-winding certificate in `TopologicalTorus.certificate`; as in the circle case, the final-domain topological groupoid theorem is unconditional, while no discrete-topology claim is made for the geometric torus loop space.

10. RELATION TO LOGIC AND REWRITING

The logical content of the construction lies in the boundary between syntax and semantics. A scoped derivation is a finite proof object built from named rules and structural constructors. Soundness is an induction principle on that derivation. Completeness is a separate metatheorem saying that the chosen syntax presents all semantic homotopies. This separation is exactly what is lost when a global equality-normalization procedure is used as a surrogate for a presentation.

Theorem 10.1 (initial semantic factorization). *Let $\mathcal{H}$ be a topological groupoid over X , and let*

$$\varphi : T \longrightarrow \mathcal{H}_1$$

be continuous. Assume that φ preserves endpoints, reflexive traces, reversal, and concatenation, and that every named scoped rewrite is sent to an equality in $\mathcal{H}_1$. Then there is a unique continuous map

$$\bar{\varphi} : G_{\mathcal{P}} \longrightarrow \mathcal{H}_1$$

such that $\bar{\varphi}q = \varphi$. It preserves all groupoid operations, with composition interpreted on the final composable domain.

Proof. The rewrite assumptions say exactly that φ is constant on the scoped setoid. The quotient universal property therefore defines $\bar{\varphi}$ and gives its uniqueness. Since q is a quotient map and φ is continuous, the quotient criterion gives continuity. The operation-preservation equations hold on representatives by assumption and descend through quotient extensionality. On the final composable domain the same argument uses the quotient of the explicit composable carrier, so no product-quotient premise is introduced. $\square$

The construction is compatible with the standard algebra of rewriting [CR36, New42, KB70, Squ87, GM09]: congruence, symmetry, transitivity, and critical-pair reasoning can be added as named presentation rules. Termination or confluence is not assumed by the topological construction. When they are used to construct a normalizer, the normal-form completeness criterion above specifies the two additional semantic checks that must be proved: geometric separation and scoped equality of equal normal forms. The finite-generator circle and torus reductions discharge those checks explicitly.

The ambient semantics is deliberately modest. The theorem uses ordinary propositional equality, continuous interval paths, and endpoint-fixed homotopies. It does not claim a HoTT identity type, univalence, or a proof-relevant equality groupoid. This scope makes the comparison suitable for a proof-irrelevant dependent

type theory while retaining explicit computational traces at the presented level [Lan98, Bro06].

11. FORMALIZATION BOUNDARY AND ARTIFACT

The mathematical statements above are independent of the implementation. A Lean 4.24.0 development with Mathlib 4.24.0 checks the construction in the seven scoped-rewrite modules under `ComputationalPaths/Path/Topology/ScopedGeometricRewrite*.lean`. The genuine torus certificate is in `ComputationalPaths/Path/Topology/TopologicalTorusScoped.lean`. The root import hub is `ComputationalPaths.lean`. The principal declaration layers are listed in Table 2.

Mathematical layer	Checked implementation declarations
Presentation and soundness	<code>ScopedGeometricRewritePresentation</code> , <code>ScopedRwEq</code> , <code>ScopedRwEq.sound</code>
Quotient and final domain	<code>scopedEquivalent</code> , <code>ScopedClass</code> , <code>ScopedComposableClass</code> , <code>scopedCompositionOnStrong</code>
Groupoid and topology	<code>ScopedStrongComposablePair</code> , <code>scopedFinalTopologicalGroupoidCertificate</code>
Ordinary-pullback compatibility	<code>scopedProductCompatibility_iff_four_way</code> , <code>scopedProductCompatibility_of_open_pair_map</code> , <code>scopedProductCompatibility_of_compact_final_t2</code> , <code>scopedProductCompatibility_of_discrete_arrow_and_final_domain</code>
Comparison and completeness	<code>toGeometricClass</code> , <code>toGeometricStrongPair</code> , <code>ComparisonFunctorCertificate</code> , <code>comparisonHomeomorph_of_complete</code> , <code>ScopedGeometricNormalFormCertificate</code> , <code>geometricCompleteness_of_normalForm</code>
Universal completion	<code>universalPresentation</code> , <code>universalScopedEquivalent_iff_totalEquivalent</code> , <code>universalHomeomorph</code>
Fundamental-groupoid bridge	<code>RealizedFundamentalArrow</code> , <code>RealizedFundamentalGroupoidCertificate</code> , <code>universalRealizedFundamentalArrowHomeomorph</code>
Circle	<code>circleLoopRule</code> , <code>circleTrace_normalizes</code> , <code>circleLoopEquivInt</code> , <code>circleBasedNormalFormCertificate</code> , <code>circleFinalTopologicalGroupoidCertificate</code>
Actual topological torus	<code>TopologicalTorus.equivIntProd</code> , <code>TopologicalTorus.standardLoop_homotopic</code> , <code>TopologicalTorus.certificate</code>

TABLE 2. Theorem-to-declaration map for the artifact.

The realized fundamental-groupoid bridge identifies the quotient of realized open paths with the range of the endpoint-and-homotopy code in Mathlib’s fundamental groupoid. Its certificate includes explicit carrier membership for identities, reversal, and composition. The comparison certificate also contains the continuous strong-pair

map and final-composition equation, so the formalized comparison is a groupoid morphism on the same final domain as the mathematical theorem.

The root reproducibility command is `lakebuild`; the focused module commands are listed in the artifact README and can be run independently.

The checked modules contain no unfinished proofs and no custom axiom declarations. Declaration-level axiom inspection reports only Lean’s standard logical and quotient principles, namely propositional extensionality, classical choice, and quotient soundness. The Lean artifact is evidence for the stated theorem package and complements the authors’ current Lean preprint [dMU21, RdOdQdV25, RdVdQdO25a]; it is not used as a substitute for the mathematical definitions or arguments in the main text.

12. CONCLUSION

Scoped computational paths provide a useful middle level between raw path syntax and semantic fundamental-groupoid arrows. The final-domain construction is the key topological device: it gives a fully unconditional groupoid multiplication while exposing, rather than hiding, the precise product-quotient condition needed for the ordinary pullback topology.

The universal presentation shows that this is not merely a bookkeeping construction: ordinary quotient-topologized fundamental groupoids arise as the geometric completion of a scoped trace system. The compact-Hausdorff and discrete finite-presentation theorems give a positive ordinary-pullback class; the effective normal-form criterion turns concrete reductions into geometric completeness. The finite-generator circle and genuine torus calculations then show that the framework retains nontrivial topological information without silently identifying scoped syntax with ambient equality.

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